

Formulation

Problem definition

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Optimization: principles and algorithms

Problem definition

We provide a generic definition of an optimization, and discuss the types of solutions that can be found.

Problem definition

$$\min_{x \in \mathbb{R}^n} f(x) \quad f : \mathbb{R}^n \rightarrow \mathbb{R}, n > 0$$

subject to

$$h(x) = 0 \quad h : \mathbb{R}^n \rightarrow \mathbb{R}^m, m \geq 0$$

$$g(x) \leq 0 \quad g : \mathbb{R}^n \rightarrow \mathbb{R}^p, p \geq 0$$

and

$$x \in X \quad X \text{ convex}$$

Feasible set

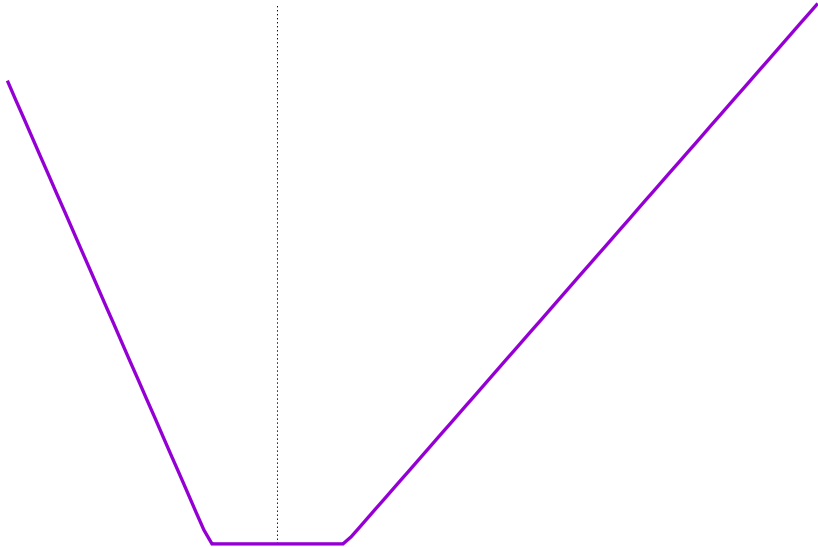
$$Y = \{x \in \mathbb{R}^n \mid h(x) = 0, g(x) \leq 0 \text{ and } x \in X\}$$

$$\min_{x \in \mathbb{R}^n} f(x)$$

subject to

$$x \in Y$$

Global optimum

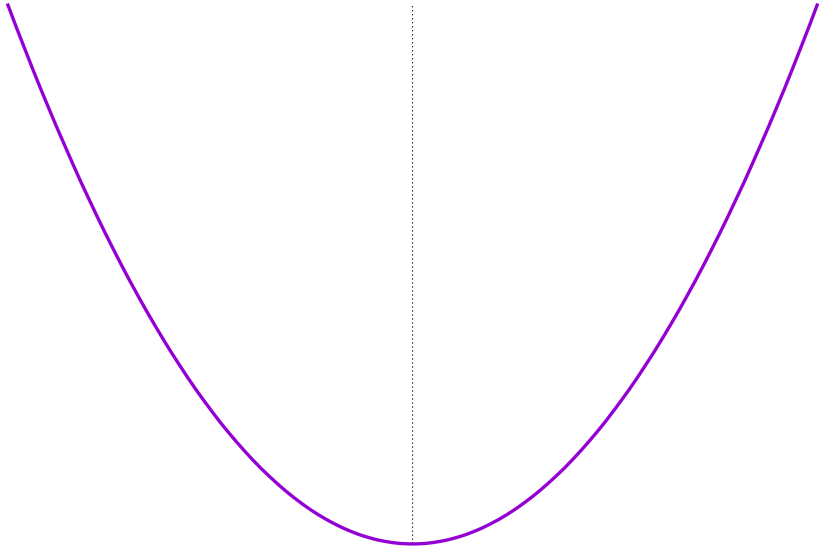


Global minimum

x^* is a global minimum if

$$f(x^*) \leq f(x), \quad \forall x \in Y.$$

Strict global optimum

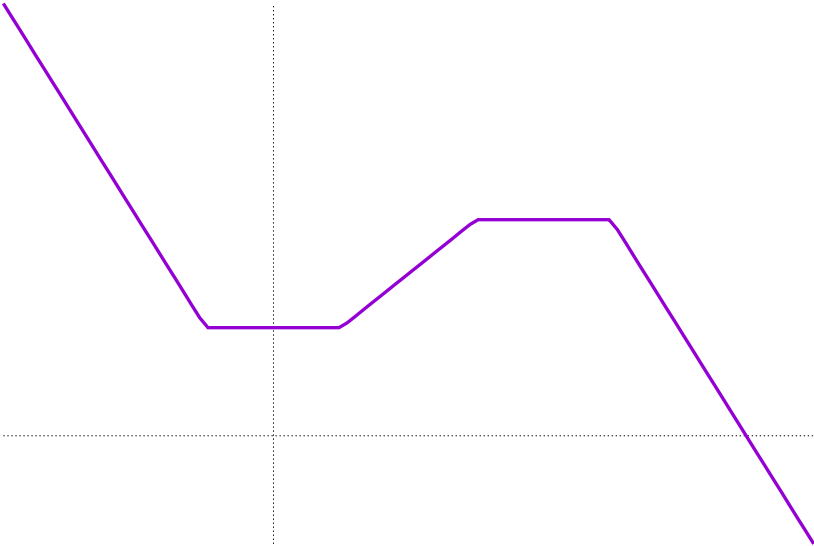


Strict global minimum

x^* is a strict global minimum if

$$f(x^*) < f(x), \quad \forall x \in Y.$$

Local optimum

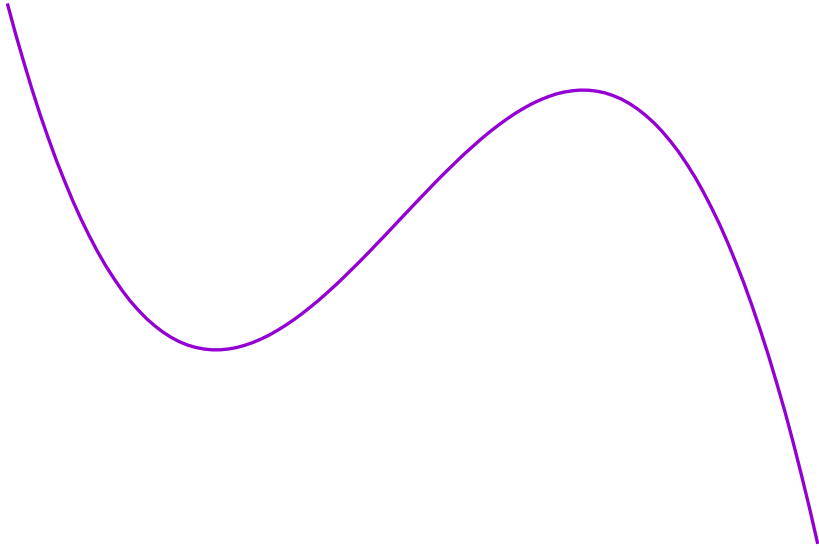


Local minimum

x^* is a local minimum if there exists $\varepsilon > 0$ such that

$$f(x^*) \leq f(x), \quad \forall x \in Y \text{ such that } \|x - x^*\| < \varepsilon.$$

Strict local optimum



Strict local minimum

x^* is a local minimum if there exists $\varepsilon > 0$ such that

$$f(x^*) < f(x), \quad \forall x \in Y \text{ such that } \|x - x^*\| < \varepsilon.$$

Summary

In the general case, only local optima will be discovered In some specific cases, namely when the function and the feasible set are convex, a local optimum is also a global one